Embedded Systems Development

Exercise 5

1. Design a "reaction timer" using ATmega328P microcontroller and give your design in a schematic diagram and a wiring diagram.

   The "reaction timer" is an embedded system that measures the user's reaction time to a signal and displays it in a scale of three LEDs. When user presses the "ready" pushbutton on the system, the "reaction timer" turns off all LEDs in the system and waits for a random amount of time between 1 and 10 seconds. Upon the expiration of this waiting time, the "reaction timer" turns on a blue LED and signals to the user. The user then presses the "react" pushbutton on the system as quickly as possible to react to the signal. The "reaction timer" then turns off the blue LED used to signal and measures the time elapsed between the signal and the user's reaction to the signal. If the user's reaction time to signal is less than one second, greater than one second but less than two seconds or greater than two seconds, the "reaction timer" turns on a green, orange or a red LED, respectively to display the user's performance.

   Schematic Diagram

Wiring Diagram

2. If the microcontroller is clocked at 16MHz and timer prescale is 1024, what is the initial value required to configure a 16-bit timer to overflow in one second?

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3. Write the required firmware for the "reaction timer" in C.
Hint: Use *srand()* function defined in *stdlib.h* and an 8-bit timer to seed the random number generator with a fairly random seed, *rand()* function defined in the same header file to generate random numbers and a 16-bit timer with a numeric variable to measure the waiting time to signal as well as the user's reaction time.

4. Modify the firmware you provided for (3) above using a floating analog input as a fairly random seed to seed the random number generator.

5. Rewrite the firmware you provided for (4) above using wrapper functions in Arduino IDE.

6. Compare the two firmware implementations that you provided for (4) and (5) above.